timer: Waveform of the Failing Run

PHASE	TEST	STATUS
Phase 2	**timer**	**WAVEFORM**

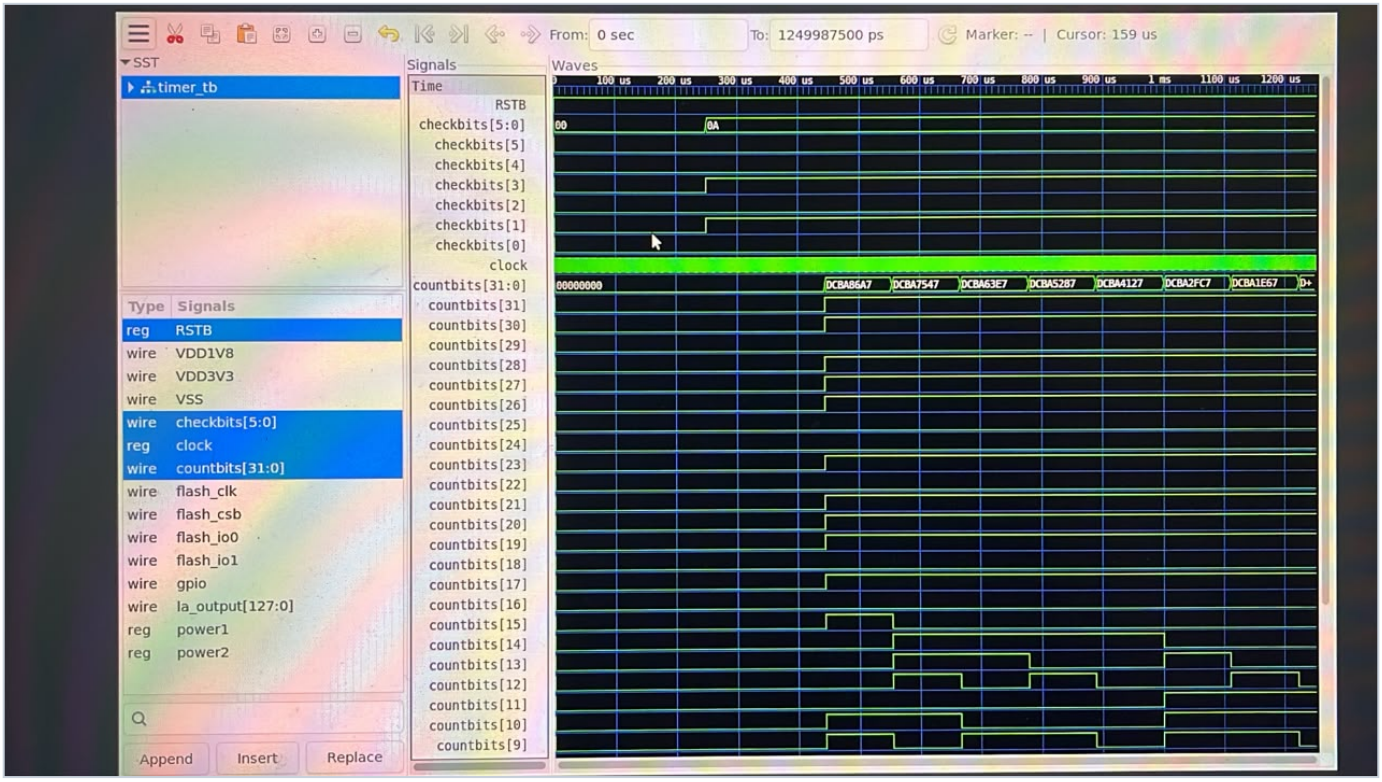


Figure 7. RTL-timer.vcd with clock, RSTB, checkbits and countbits

What this shows

- `clock` runs and RSTB is released: not a clock or reset problem.
- `checkbits` reaches `0A` (the firmware's start marker) and stays there: the next checkpoint was never reached.
- `countbits` keeps changing (`DCBA86A7, DCBA7547, DCBA63E7, ...`, counting down): the timer works and the CPU is alive.

> **Conclusion.** The firmware was running correctly but had not finished when the limit expired. Hypothesis: the time limit is slightly too short.